

Section 1: Given $f(x) = -3x^3$ and $g(x) = x^2 - 5$ find the following.

1. $f(g(2))$

$$g(2) = 2^2 - 5 = -1$$

$$f(-1) = -3(-1)^3 = 3$$

2. $[g \circ f](2)$

$$f(2) = -3(2^3) = -24$$

$$g(-24) = (-24)^2 - 5 = 571$$

Section 2: Find $[f \circ g](x)$ and $[g \circ f](x)$.

3. $f(x) = \frac{1}{2}x - 4$

$$g(x) = 4x + 24$$

$$\frac{1}{2}(4x + 24) - 4 \quad f(g(x))$$

$$2x + 12 - 4$$

$$= 2x + 8$$

$$4\left(\frac{1}{2}x - 4\right) + 24 \quad g(f(x))$$

$$2x - 16 + 24$$

$$= 2x + 8$$

4. $f(x) = 5x$

$$g(x) = 2x^3 + 3x^2 + x - 1$$

$$f(g(x)) = 5(2x^3 + 3x^2 + x - 1)$$

$$= 10x^3 + 15x^2 + 5x - 5$$

$$g(f(x)) = 2(5x)^3 + 3(5x)^2 + 5x - 1$$

$$= 2(125x^3) + 3(25x^2) + 5x - 1$$

$$= 250x^3 + 75x^2 + 5x - 1$$

Section 3: Use $f(g(x))$ and $g(f(x))$ to determine if f and g are inverses. Show all work!

5. $f(x) = 3x^2 - 11$

$$g(x) = \sqrt{\frac{x+11}{3}}$$

$$f(g(x)) = 3\left(\sqrt{\frac{x+11}{3}}\right)^2 - 11$$

$$= (x+11) - 11$$

$$= x$$

$$g(f(x)) = \sqrt{\frac{3x^2 - 11 + 11}{3}}$$

$$= \sqrt{\frac{3x^2}{3}}$$

$$= \sqrt{x^2} = x$$

They are not inverses!

Section 4: Find $f^{-1}(x)$ for the following functions

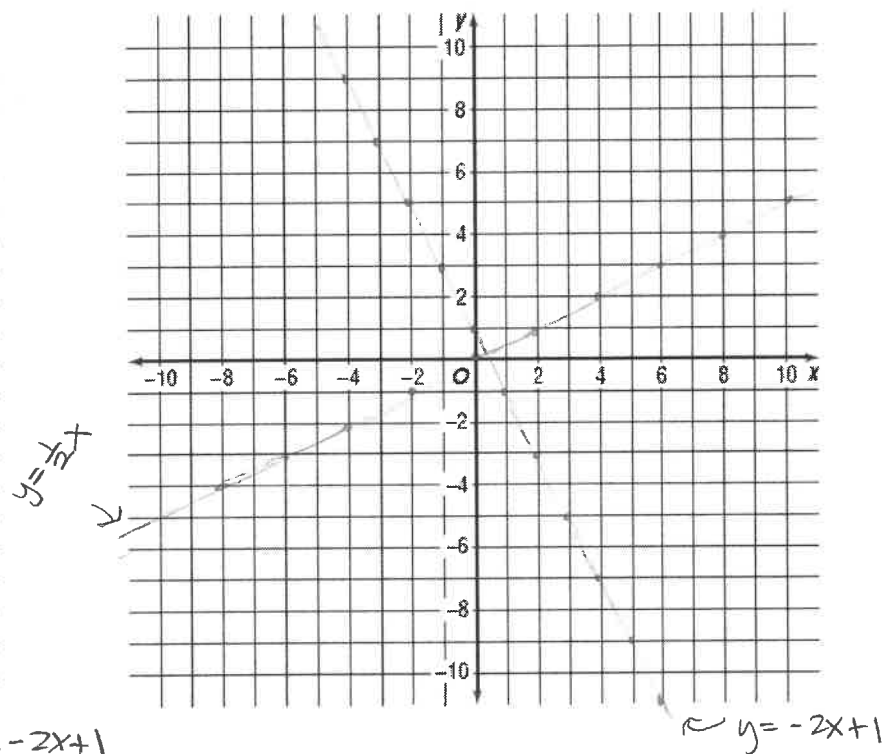
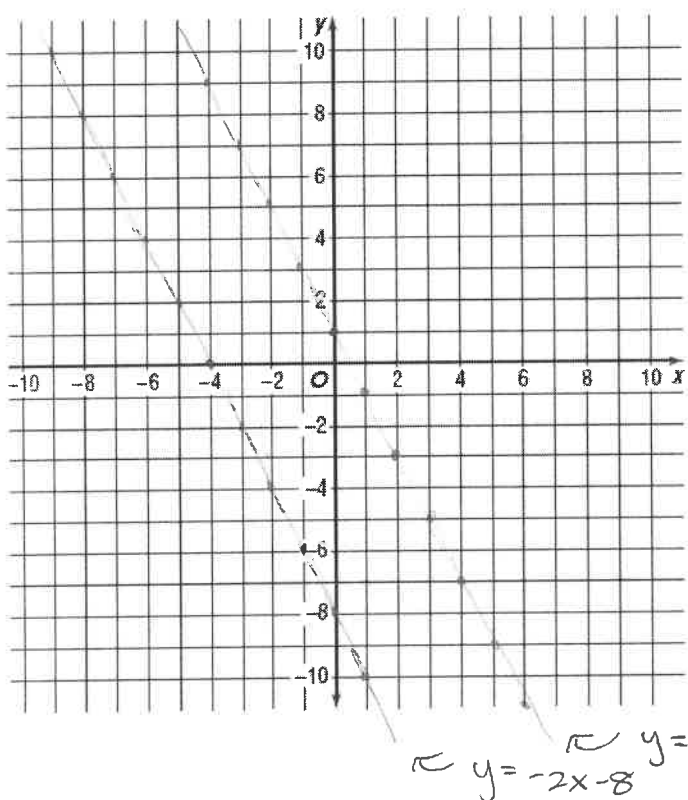
6. $f(x) = x^2 + 7$
 $x = y^2 + 7$
 $x - 7 = y^2$
 $\pm\sqrt{x-7} = y$ or
 $f^{-1}(x) = \pm\sqrt{x-7}$

7. $f(x) = \sqrt[3]{\frac{x-2}{5}}$
 $x = \sqrt[3]{\frac{y-2}{5}}$
 $x^3 = \frac{y-2}{5}$
 $5x^3 = y-2$
 $5x^3 + 2 = y$ or $f^{-1}(x) = 5x^3 + 2$

1. The slope of a horizontal line is zero.
2. Two lines are perpendicular if they intersect to form right angles.

Graph a line parallel to $y = -2x + 1$ that passes through $(-4, 0)$

Graph a line perpendicular to $y = -2x + 1$ that passes through $(0, 0)$



$$1) \log_2 16 - \log_3 81$$

Solve $4 - 4 = 0$

$$1. \log_9 \frac{1}{9} = x$$

$$x = -1$$

$$3. \log_9 (6x+7) = 2$$

$$6x+7 = 9^2$$

$$6x+7 = 81$$

$$\frac{6x}{6} = \frac{74}{6} = \frac{37}{3} = x$$

$$5. \log x = -2$$

$$10$$

$$x = 10^{-2}$$

$$x = \frac{1}{100}$$

$$7. \log_6 \frac{1}{36} = -2$$

$$\uparrow$$

$$-2$$

$$6^{-2} = \frac{1}{36}$$

$$2) \log_5 125 + \log_4 16 - \log_2 32$$

$$= 0$$

$$2. \log_3 3 + \log_2 8 = x$$

$$1 + 3 = 4$$

$$x = 4$$

$$4. \log_6 36 = 5x + 3$$

$$2 = 5x + 3$$

$$\frac{1}{2}$$

$$\frac{-1}{5} = \frac{5x}{5}$$

$$x = -\frac{1}{5}$$

$$6. \log_5 \sqrt{5} = x$$

$$x = \frac{1}{2}$$

$$8. \log_8 4 = \frac{2}{3}$$

$$8 = 2^3$$

$$4 = 2^2$$

$$2^{3x} = 2^2$$

$$\frac{3x}{3} = \frac{2}{3}$$

Express in scientific notation

0.0000456

$$4.56 \times 10^{-5}$$

Simplify:

$$\begin{aligned} & \frac{1.4 \times 10^4}{0.2 \times 10^2} \quad 4-2=2 \\ & \quad \quad \quad = 10^2 \\ & \quad \quad \quad 7 \cdot 10^2 = 700 \\ & \Rightarrow 7 \rightarrow (2 \times 10^4)(4 \times 10^{-5}) \\ & \quad \quad \quad = 8 \times 10^{-1} \\ & \quad \quad \quad = \frac{8}{10} \\ & \quad \quad \quad = \left(\frac{4}{5}\right) \end{aligned}$$

78,000,000,

$$7.8 \times 10^7$$

$$\begin{aligned} & \quad \quad \quad = 3 \\ & \quad \quad \quad \downarrow \\ & \quad \quad \quad \left(\frac{3.3 \times 10^{-3}}{1.1 \times 10^{-5}} \right) \quad \downarrow \quad -3 - (-5) = 2 \\ & \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad 3 \cdot 10^2 = 300 \\ & \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \downarrow \\ & \quad \quad \quad (7 \times 10^3)(3 \times 10^{-1}) \\ & \quad \quad \quad = 21 \times 10^2 \\ & \quad \quad \quad = 2100 \end{aligned}$$

Quiz Graph Behavior

$$1. \quad f(x) = \frac{-x^2 + x}{x-1} \quad \frac{-x(x-1)}{(x-1)}$$

Y - Intercept (0,0)

Vertical Asymptotes none

Horizontal Asymptote none

$$3. \quad f(x) = \frac{2x^2 + x - 1}{x^2 - 1} \quad \frac{(2x-1)(x+1)}{(x-1)(x+1)} = \frac{-1}{-1}$$

Y - Intercept (0,1)

Vertical Asymptotes x=1

Horizontal Asymptote y=2

$$5. \quad f(x) = \frac{x+2}{x-3}$$

Y - Intercept (0, -2/3)

Vertical Asymptotes x=3

Horizontal Asymptote y=1

$$7. \quad f(x) = \frac{-x^2 + 2}{x+2}$$

Y - Intercept (0,1)

Vertical Asymptotes x=-2

Horizontal Asymptote none

$$2. \quad f(x) = \frac{x^2 + 2}{x-1} \quad \frac{0+2}{-1} = -2$$

Y - Intercept (0,-2)

Vertical Asymptotes x=1

Horizontal Asymptote none

$$4. \quad f(x) = \frac{2}{x^2 - 16}$$

Y - Intercept (0, -1/8)

Vertical Asymptotes x=±4

Horizontal Asymptote y=0

$$6. \quad f(x) = \frac{x^2 + 2}{x-1}$$

Y - Intercept (0,-2)

Vertical Asymptotes x=1

Horizontal Asymptote none

$$8. \quad f(x) = \frac{2}{x^2 - 4}$$

Y - Intercept (0, -1/2)

Vertical Asymptotes x=±2

Horizontal Asymptote y=0

$$9. \quad f(x) = \frac{2x^2 + x - 1}{x^2 - 1} \quad \frac{(2x-1)(x+1)}{(x+1)(x-1)}$$

Y - Intercept (0, 1)

Vertical Asymptotes $x = 1$

Horizontal Asymptote $y = 2$

$$10. \quad f(x) = \frac{x+4}{x^2+9x+20} \quad \frac{(x+4)}{(x+5)(x+4)}$$

Y - Intercept (0, $\frac{1}{5}$)

Vertical Asymptotes $x = -5$

Horizontal Asymptote $y = 0$

$$11. \quad f(x) = \frac{x^2 - 4x + 3}{x - 1} \quad \frac{(x-3)(x-1)}{(x-1)}$$

Y - Intercept (0, -3)

Vertical Asymptotes none

Horizontal Asymptote none

$$12. \quad f(x) = \frac{4x+3}{3x-5} \quad \frac{3x-5}{3x-5}$$

Y - Intercept (0, $-\frac{3}{5}$)

Vertical Asymptotes $x = \frac{5}{3}$

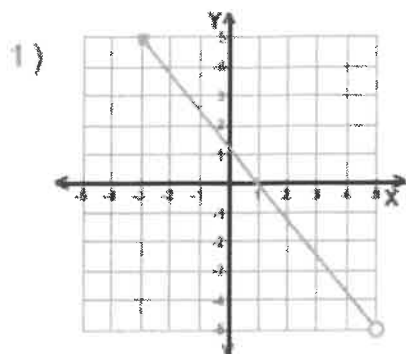
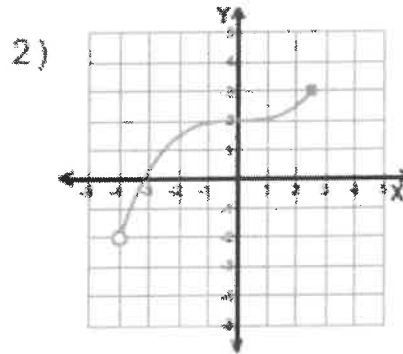
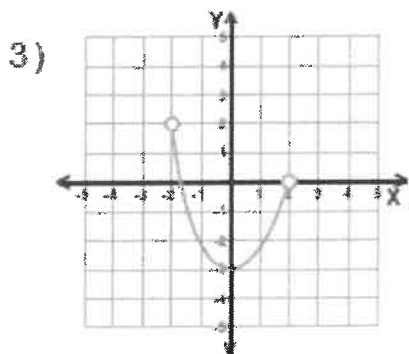
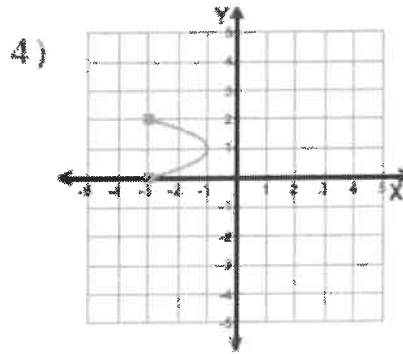
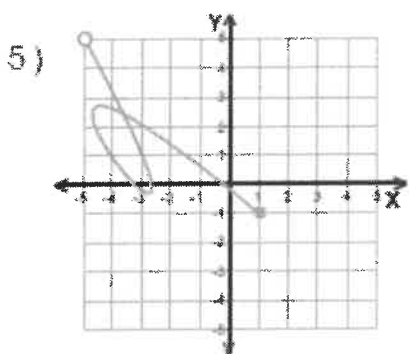
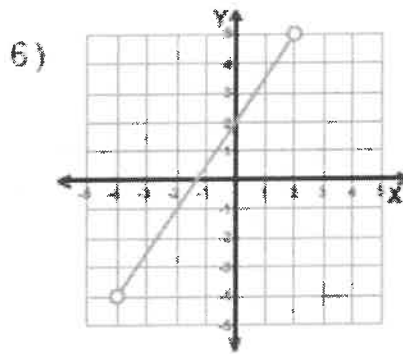
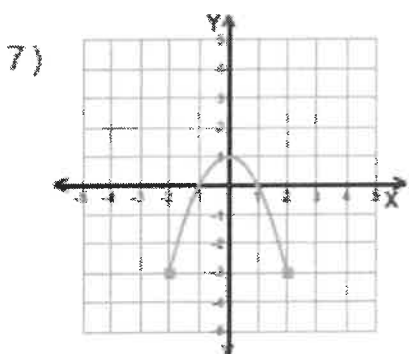
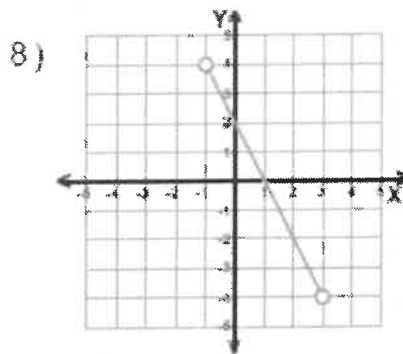
Horizontal Asymptote $y = \frac{4}{3}$

Write a rational function that has a horizontal asymptote at $y = 2$.

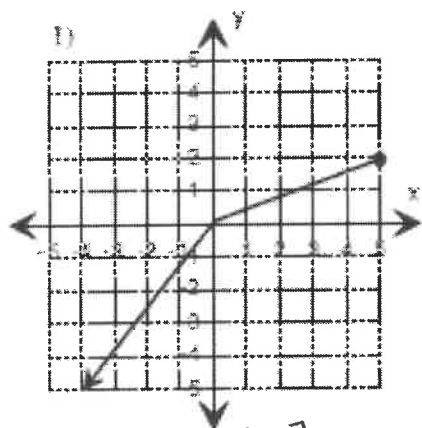
$$f(x) = \frac{2x^2 + 5}{x^2 - 1}$$

Design a rational function that has a vertical asymptote at $x = -7$.

$$f(x) = \frac{x^2 - 1}{x + 7}$$

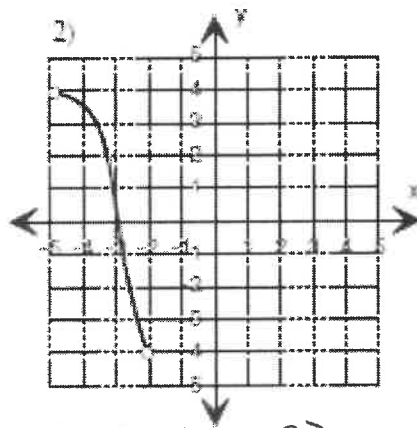
Domain: $[-3, 5]$ Range: $(-5, 5]$ Domain: $[-4, 2\frac{1}{2}]$ Range: $(-2, 3]$ Domain: $(-2, 2)$ Range: $[-3, 2)$ Domain: $[-3, -1]$ Range: $[0, 2]$ Domain: $(-5, 1]$ Range: $[-1, 5)$ Domain: $(-4, 2)$ Range: $(-4, 5)$ Domain: $[-2, 2]$ Range: $[-3, 1]$ Domain: $(-1, 3)$ Range: $(-4, 4)$

Find the Domain and Range for each graph.



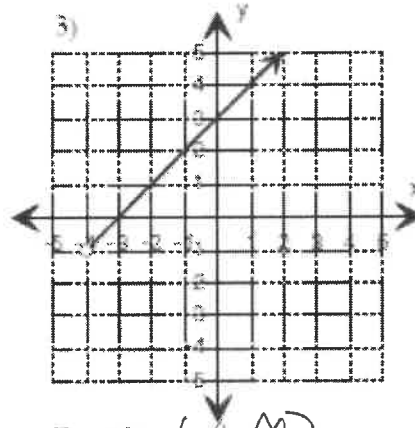
Domain: $(-\infty, 5]$

Range: $(-\infty, 2]$



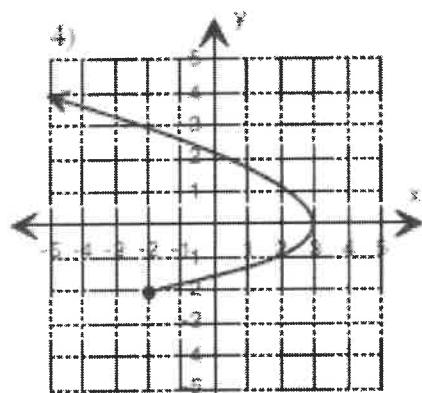
Domain: $(-5, -2)$

Range: $(-4, 4)$



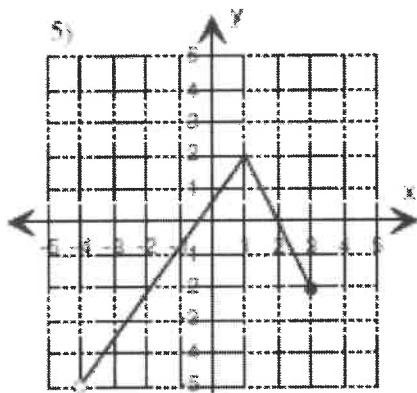
Domain: $(-4, \infty)$

Range: $(-\infty, \infty)$



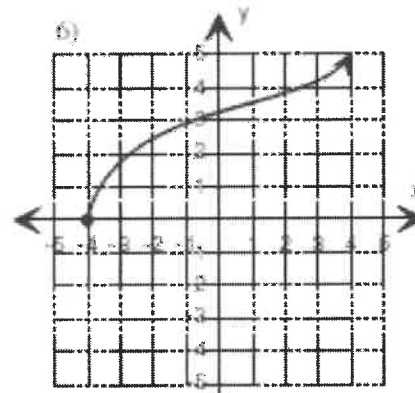
Domain: $(-\infty, 3]$

Range: $[-2, \infty)$



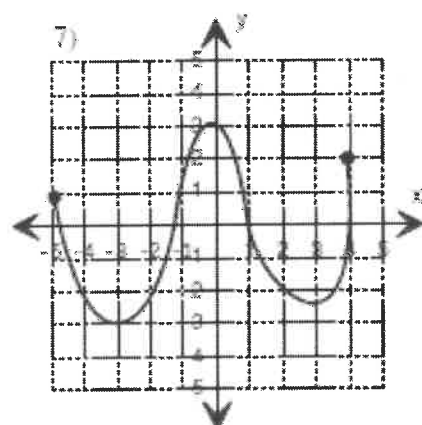
Domain: $(-4, 3]$

Range: $(-5, 2]$



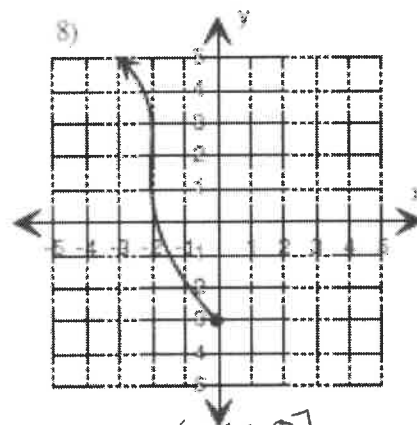
Domain: $[-4, \infty)$

Range: $[0, \infty)$



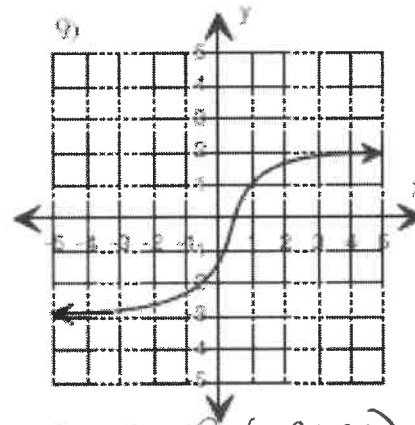
Domain: $[-5, 4]$

Range: $[-3, 3]$



Domain: $(-\infty, 0]$

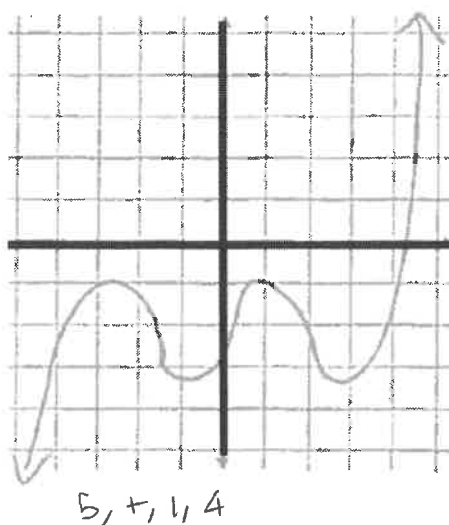
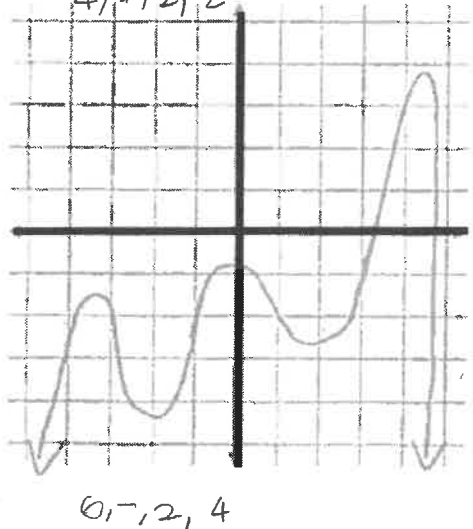
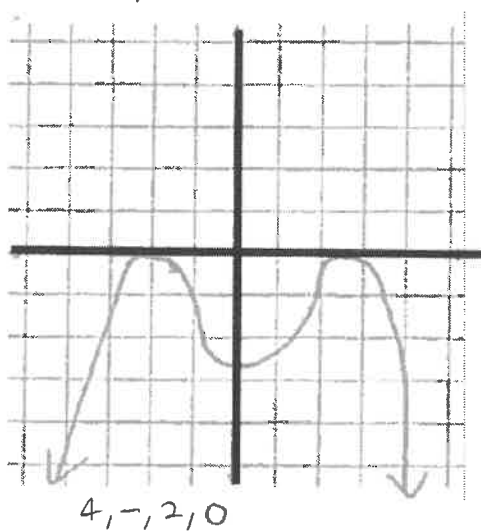
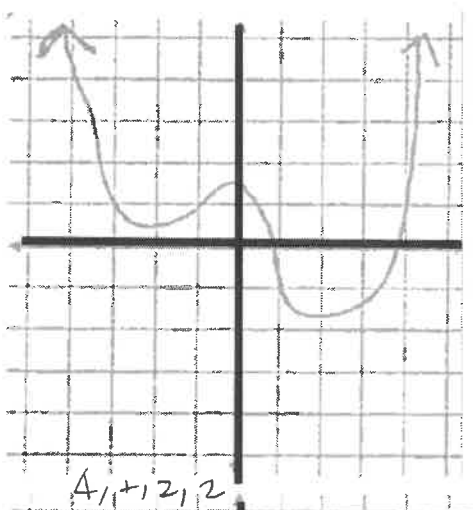
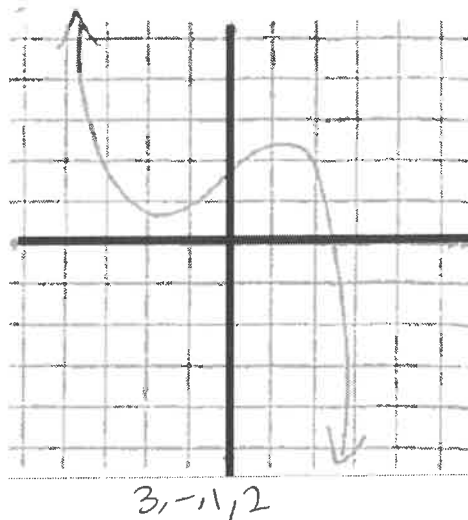
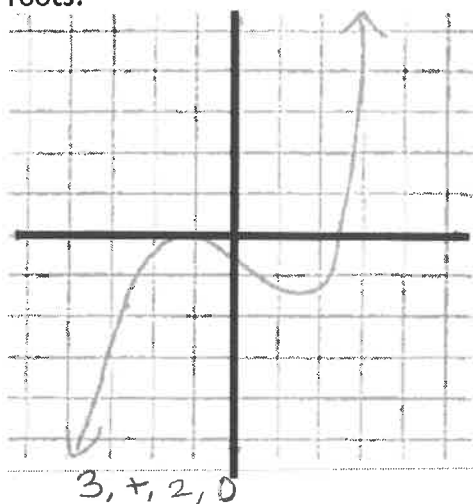
Range: $[-3, \infty)$

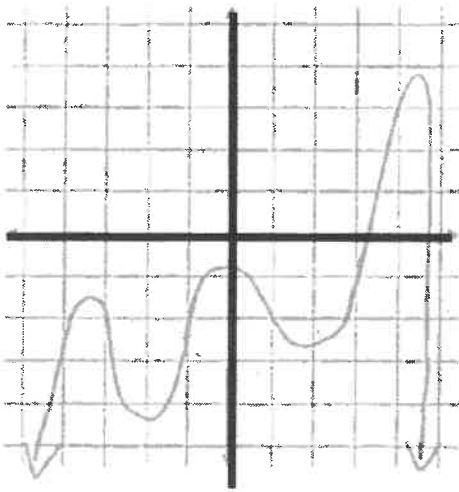


Domain: $\mathbb{R}(-\infty, \infty)$

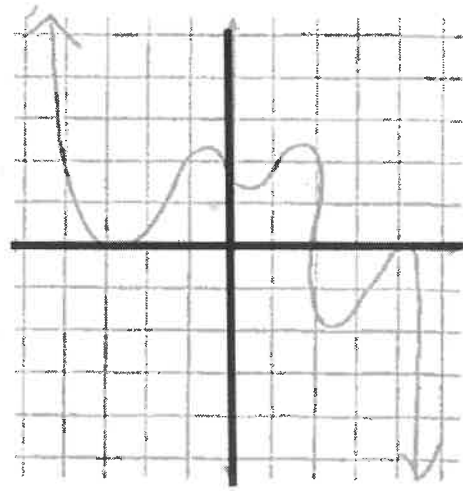
Range: $\mathbb{R}(-\infty, \infty)$

For each graph, state the degree, sign of leading coefficient, number of real roots, and number of complex roots.

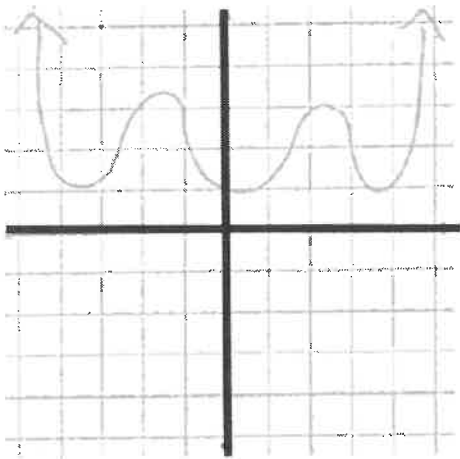




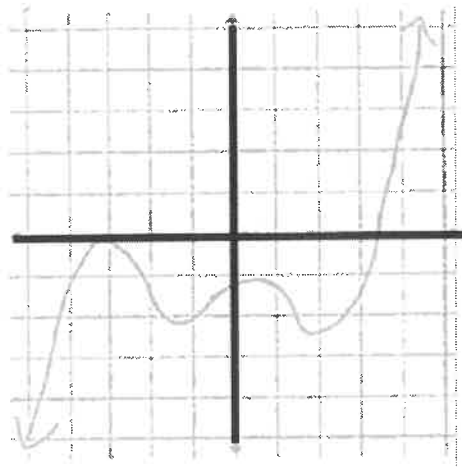
$6, -1.2, 4$



$7, -1.3, 2$



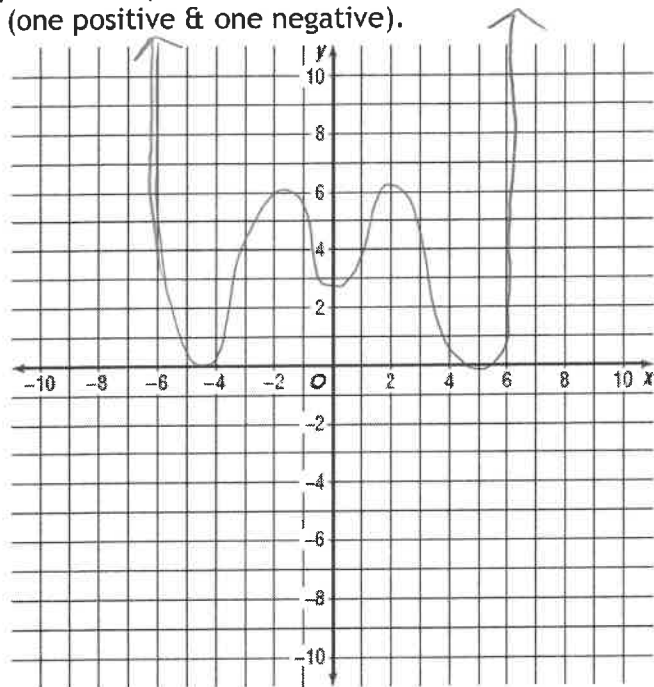
$6, +, 0, 0$



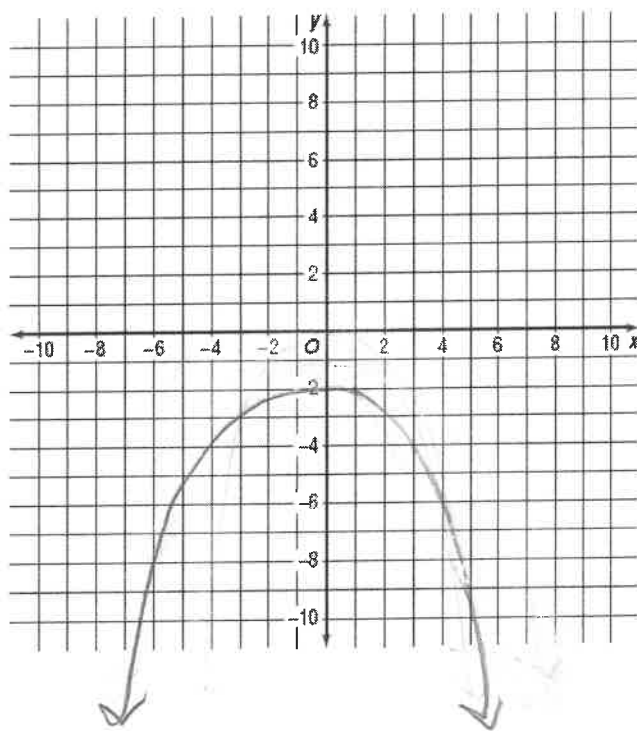
$5, +, 2, 2$

Graphing

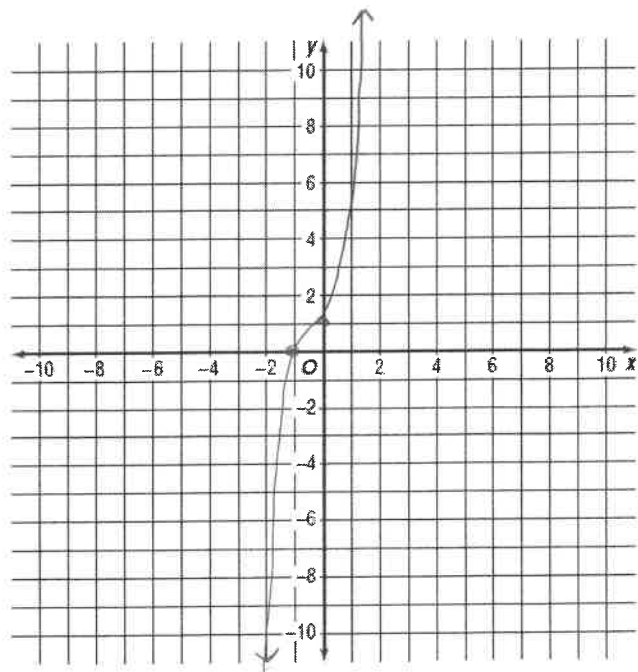
1. Draw a possible graph of a positive 6th degree function that contains exactly one pair of complex roots and two double roots (one positive & one negative).



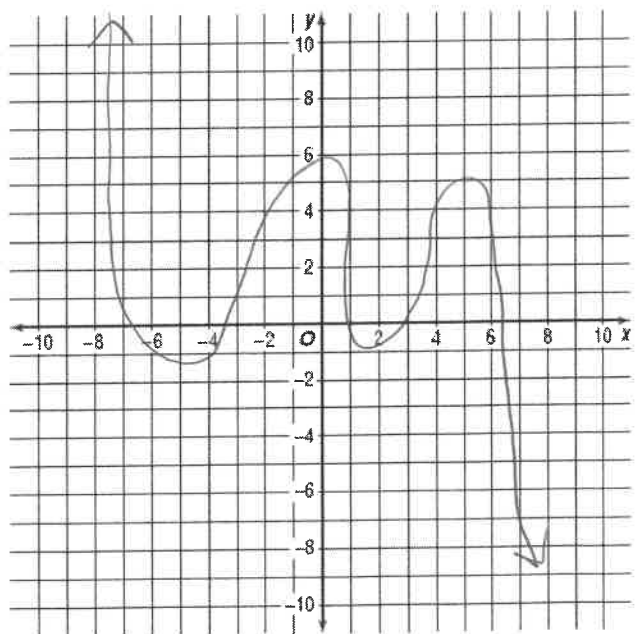
2. Draw a possible graph for $f(x) = -a_8x^8 + a_0$ with no real roots.



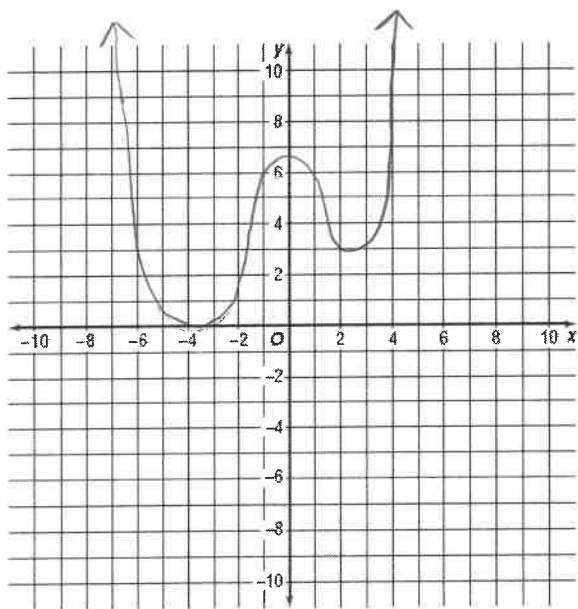
3. Sketch a possible graph for $f(x) = x^3 + x^2 + x + 1$



4. Draw a possible graph of a negative fifth degree polynomial with exactly three positive real roots and two negative real roots.

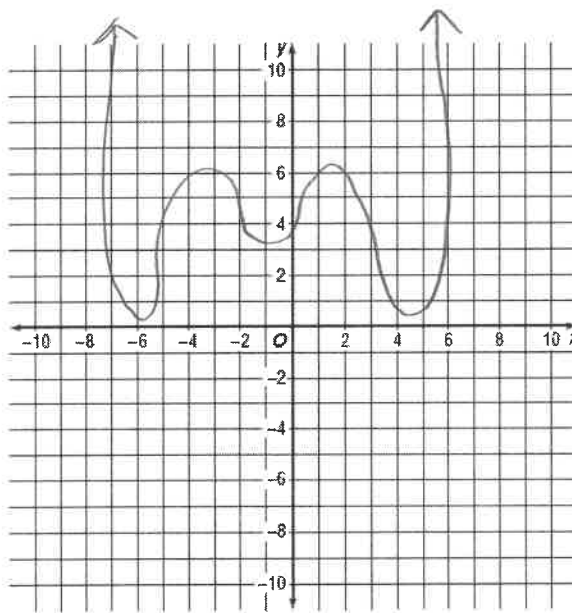


1. Draw a graph of a positive quartic function that contains exactly one double root.

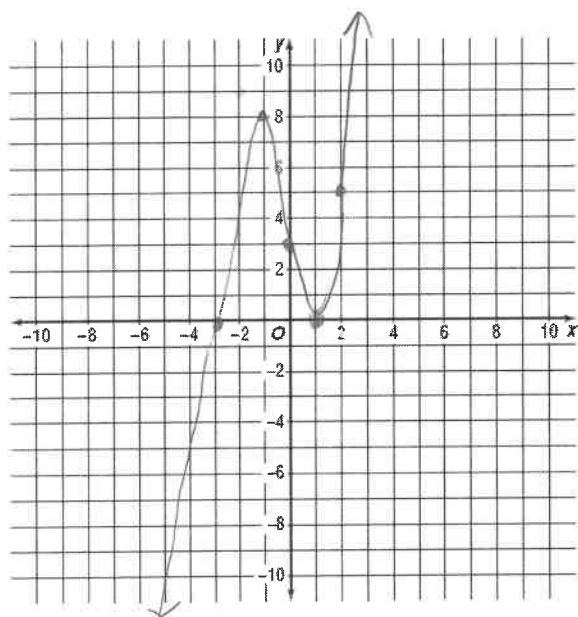


2. Draw a possible graph for

$$f(x) = a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0.$$



3. Graph $f(x) = (x-1)^2(x+3)$ by finding x and y intercepts.



$$y \text{ int} = (0, 3)$$

$$x \text{ int} = (-3, 0) \quad (1, 0)$$